Section 1: These questions and solutions are designed to assess understanding of key concepts in auditing Windows operating systems, including account policies, password complexities, audit log management, patch management, and compliance measurement

Question 1: Calculating Account Lockout Duration Impact

A company's security policy mandates that user accounts should be locked out after 5 unsuccessful login attempts within 15 minutes. The account lockout duration is set to 30 minutes.

Scenario: An attacker initiates a brute-force attack, attempting to guess a user's password by trying 4 different passwords every 10 minutes.

Calculate: The total time (in hours) it would take for the attacker to make 1000 password attempts under this policy.

Solution:

- **Attempts before lockout:** The attacker makes 4 attempts every 10 minutes. After 5 unsuccessful attempts within 15 minutes, the account is locked for 30 minutes.
- **Time calculation:**
  - First 5 attempts: 10 minutes (4 attempts) + 2.5 minutes (1 attempt) = 12.5 minutes.
  - Account lockout: 30 minutes.
  - Total cycle time for 5 attempts: 12.5 minutes + 30 minutes = 42.5 minutes.
- **Total cycles needed for 1000 attempts:** 1000 attempts ÷ 5 attempts per cycle = 200 cycles.
- **Total time:** 200 cycles × 42.5 minutes per cycle = 8500 minutes.
- **Convert to hours:** 8500 minutes ÷ 60 minutes per hour ≈ 141.67 hours.

Answer: It would take approximately 141.67 hours for the attacker to make 1000 password attempts under this account lockout policy.

Question 2: Evaluating Password Policy Strength

A Windows operating system enforces a password policy requiring passwords to be at least 8 characters long, including at least one uppercase letter, one lowercase letter, one digit, and one special character.

Calculate: The total number of possible unique passwords under this policy.

Solution:

- **Character set size:**
  - Uppercase letters: 26
  - Lowercase letters: 26
  - Digits: 10
  - Special characters (assuming standard US keyboard): 32
  - Total: 26 + 26 + 10 + 32 = 94 characters.
- **Password length:** 8 characters.
- **Total possible combinations:** 94^8 (94 raised to the power of 8).
- **Calculation:** 94^8 ≈ 6.095 × 10^15.

Answer: There are approximately 6.095 quadrillion possible unique passwords under this policy.

Question 3: Assessing Audit Log Storage Requirements

An organization configures its Windows servers to log audit events, generating an average of 5000 events per day per server. Each log entry is approximately 500 bytes in size.

Calculate: The total storage (in gigabytes) required to retain audit logs for 90 days across 50 servers.
Solution:

- **Daily log size per server:** 5000 events × 500 bytes = 2,500,000 bytes = 2.5 MB.
- **Total log size per server for 90 days:** 2.5 MB/day × 90 days = 225 MB.
- **Total log size for 50 servers:** 225 MB × 50 servers = 11,250 MB.
- **Convert to gigabytes:** 11,250 MB ÷ 1024 MB/GB ≈ 10.99 GB.

Answer: Approximately 10.99 GB of storage is required to retain audit logs for 90 days across 50 servers.
Question 4: Determining Patch Deployment Timeframe
A company needs to deploy a critical security patch to 200 Windows servers. Each patch installation takes approximately 15 minutes, and the IT team can work on 10 servers simultaneously.
Calculate: The minimum time (in hours) required to patch all servers.
Solution:

- **Total batches:** 200 servers ÷ 10 servers per batch = 20 batches.
- **Time per batch:** 15 minutes.
- **Total time:** 20 batches × 15 minutes = 300 minutes.
- **Convert to hours:** 300 minutes ÷ 60 minutes per hour = 5 hours.

Answer: It would take a minimum of 5 hours to patch all 200 servers.
Question 5: Calculating Security Update Compliance Rate
During an audit of 150 Windows workstations, it was found that 120 had all security updates installed, while the remaining had varying degrees of missing updates.
Calculate: The compliance rate as a percentage.
Solution:

- **Compliance rate:** (Number of compliant workstations ÷ Total number of workstations) × 100
- **Calculation:** (120 ÷ 150) × 100 = 80%.

Answer: The security update compliance rate is 80%.

Section 2: These questions are designed to test the student's understanding of auditing metrics, percentages, and the interpretation of Windows security log data.

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1. Question: User Account Auditing
A company is auditing its Windows Server 2016 environment. They have enabled user account auditing to track login attempts. The server logs show that the following user login attempts occurred in the past 24 hours:

- 15 successful logins
- 7 failed logins
- 2 invalid username attempts
- 3 lockout events (users attempting after exceeding failed login attempts)

Calculate the following:

- **Success rate of login attempts**
- **Failure rate of login attempts**
- **Percentage of lockout events relative to total login attempts**

Solution:

- Total login attempts = 15 (successful) + 7 (failed) + 2 (invalid usernames) = 24 attempts
- **Success rate = (Successful logins / Total login attempts) × 100** = (15 / 24) × 100 = **62.5%**
- **Failure rate = (Failed logins + Invalid username attempts / Total login attempts) × 100** = (7 + 2) / 24 × 100 = **37.5%**
- **Percentage of lockout events = (Lockout events / Total login attempts) × 100** = 3 / 24 × 100 = **12.5%**

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2. Question: Failed Login Attempt Analysis

In the Windows event log, a report has been generated for failed login attempts in a week. The report shows the following:

- Total failed login attempts: 120
- 35 of these failed attempts were due to incorrect passwords.
- 45 failed attempts were caused by disabled accounts.
- 15 attempts resulted in account lockout.
- 25 failed attempts were due to expired user passwords.

Calculate:

- **Percentage of failed login attempts due to incorrect passwords**
- **Percentage of failed login attempts due to disabled accounts**
- **Percentage of failed login attempts resulting in lockouts**

Solution:

- Total failed login attempts = 120
- **Percentage of incorrect passwords = (Incorrect passwords / Total failed login attempts) × 100** = (35 / 120) × 100 = **29.17%**
- **Percentage of disabled accounts = (Disabled accounts / Total failed login attempts) × 100** = (45 / 120) × 100 = **37.5%**
- **Percentage of lockouts = (Lockouts / Total failed login attempts) × 100** = (15 / 120) × 100 = **12.5%**

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3. Question: Password Expiration Auditing

A network administrator is auditing password expiration events on a Windows environment. In a 30-day period, 250 users were found to have passwords that expired. The administrator notes that the passwords of the following number of users expired at the following intervals:

- 100 users had passwords expire within 10 days of the expiration date
- 75 users had passwords expire within 11-20 days
- 50 users had passwords expire after 20 days
- 25 users were unaware of the expiration until their accounts were locked.

Calculate the following:

- **Percentage of users who expired their passwords within 10 days**
- **Percentage of users who expired their passwords within 11-20 days**
- **Percentage of users who expired their passwords after 20 days**
- **Percentage of users who did not know about the expiration**

Solution:

- Total users = 250

- **Percentage of users who expired passwords within 10 days = (Users expired within 10 days / Total users) × 100** = (100 / 250) × 100 = **40%**
- **Percentage of users who expired passwords within 11-20 days = (Users expired within 11-20 days / Total users) × 100** = (75 / 250) × 100 = **30%**
- **Percentage of users who expired passwords after 20 days = (Users expired after 20 days / Total users) × 100** = (50 / 250) × 100 = **20%**
- **Percentage of users unaware of expiration = (Users unaware / Total users) × 100** = (25 / 250) × 100 = **10%**

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4. Question: Account Lockout Policy Audit

The Windows server environment has an account lockout policy configured as follows:

- Lockout threshold: 5 failed login attempts
- Lockout duration: 15 minutes
- Reset counter after 30 minutes

In the last week, the system logged the following:

- 150 failed login attempts
- 10 accounts locked out
- 5 accounts had multiple lockouts during the week.

Calculate the following:

- **Average failed login attempts per locked-out account**
- **Percentage of lockouts out of total failed login attempts**
- **Percentage of accounts with multiple lockouts**

Solution:

- Total failed login attempts = 150
- Total lockouts = 10
- **Average failed login attempts per locked-out account = Total failed login attempts / Total lockouts** = 150 / 10 = **15 failed attempts per locked-out account**
- **Percentage of lockouts = (Lockouts / Total failed login attempts) × 100** = (10 / 150) × 100 = **6.67%**
- **Percentage of accounts with multiple lockouts = (Accounts with multiple lockouts / Total lockouts) × 100** = (5 / 10) × 100 = **50%**

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5. Question: Audit Log Data Review

A Windows system administrator is reviewing audit logs for unauthorized access attempts. The logs show:

- Total access attempts logged: 500
- Authorized access attempts: 450
- Unauthorized access attempts: 50
- Total successful access attempts: 430
- Total failed access attempts: 70

Calculate the following:

- **Percentage of successful access attempts**
- **Percentage of failed access attempts**
- **Percentage of unauthorized access attempts**

Solution:

- Total access attempts = 500

- **Percentage of successful access attempts = (Successful access attempts / Total access attempts) × 100** = (430 / 500) × 100 = **86%**
- **Percentage of failed access attempts = (Failed access attempts / Total access attempts) × 100** = (70 / 500) × 100 = **14%**
- **Percentage of unauthorized access attempts = (Unauthorized access attempts / Total access attempts) × 100** = (50 / 500) × 100 = **10%**

Section 3: These questions are designed to test the student's understanding of auditing metrics, percentages, and the interpretation of Windows Event Log Analysis, Windows Group Policy Evaluation, System Patch Management

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Question 1: Event Log Analysis

You are reviewing Windows Event Logs for signs of unauthorized login attempts. The security log shows the following sequence of events over the last 48 hours:

- Event 529: 5 failed login attempts from User1 at 3:00 AM.
- Event 529: 8 failed login attempts from User2 at 6:30 AM.
- Event 538: 2 successful logins from User1 at 10:00 AM.
- Event 529: 10 failed login attempts from User3 at 1:30 PM.

If the threshold for failed login attempts is set to 5, how many accounts would be flagged for potential account lockout?

Solution: Failed login attempts over the threshold of 5:

- User1: 5 failed attempts — no flag (exactly at the threshold)
- User2: 8 failed attempts — flagged
- User3: 10 failed attempts — flagged

Answer: 2 accounts (User2 and User3) would be flagged for potential lockout.

Question 2: Windows Group Policy Evaluation

You are reviewing a Windows Group Policy setting that applies to a company's 100 computers. The policy is configured to disable the "Guest" account on all computers. During an audit, you find that 3 computers have the "Guest" account enabled. What percentage of computers are non-compliant with the Group Policy?

Solution: Out of 100 computers, 3 are non-compliant, so the percentage of non-compliance is:
$3/100 \times 100 = 3\%$

Answer: 3% of the computers are non-compliant.

Question 3: System Patch Management

You are auditing patch management for a Windows Server. The server has installed 15 out of 20 critical patches in the last month. What percentage of the critical patches has been installed?

Solution: The percentage of installed patches is:
$15/20 \times 100 = 75\%$

Answer: 75% of the critical patches have been installed.
